

Multi-Object Detection and Persistent ID Tracking

1. Model / Detector Used

In this project, **YOLOv8 (You Only Look Once Version 8)** was used as the object detection model. YOLOv8 is a state-of-the-art real-time object detection algorithm known for its high speed and accuracy. It processes each frame of the video and detects objects such as people by generating bounding boxes along with confidence scores.

2. Tracking Algorithm Used

For multi-object tracking, the **DeepSORT (Deep Simple Online Realtime Tracking)** algorithm was used. DeepSORT extends the SORT algorithm by incorporating appearance-based features along with motion information, enabling more reliable tracking across frames.

3. Why This Combination Was Selected

The combination of YOLOv8 and Deep SORT was chosen due to the following reasons:

- **YOLOv8** provides fast and accurate object detection suitable for real-time applications
 - **Deep SORT** ensures robust tracking by maintaining identity across frames
 - The integration is widely used and well-supported in practical computer vision tasks
 - It performs well under moderate occlusion and movement scenarios
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4. How ID Consistency is Maintained

ID consistency is maintained using a combination of:

- **Motion Prediction (Kalman Filter):** Predicts the next position of objects
- **Data Association (Hungarian Algorithm):** Matches detected objects with existing tracks
- **Appearance Features (Re-identification):** Uses visual features to distinguish similar-looking objects

This multi-factor approach allows the system to maintain consistent IDs even when objects temporarily disappear or overlap.

5. Challenges Faced

Several real-world challenges were encountered during the implementation:

- **Occlusion:** Players blocking each other caused temporary tracking loss
- **Motion Blur:** Fast movement reduced detection accuracy
- **Scale Variation:** Objects appearing smaller or larger due to camera zoom

- **Similar Appearance:** Players wearing similar jerseys led to confusion in tracking
 - **Camera Movement:** Panning and zooming affected object consistency
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6. Failure Cases Observed

Some failure cases observed include:

- ID switching when two players overlap for a long duration
 - Missed detections during fast motion or low visibility
 - New IDs assigned when objects reappear after disappearing
 - Reduced accuracy in crowded scenes
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7. Possible Improvements

The following improvements can enhance system performance:

- Using a larger model like YOLOv8m or YOLOv8l for better detection accuracy
- Replacing DeepSORT with ByteTrack for improved tracking performance
- Incorporating a stronger re-identification model
- Applying video stabilization techniques
- Adding trajectory analysis and motion prediction enhancements